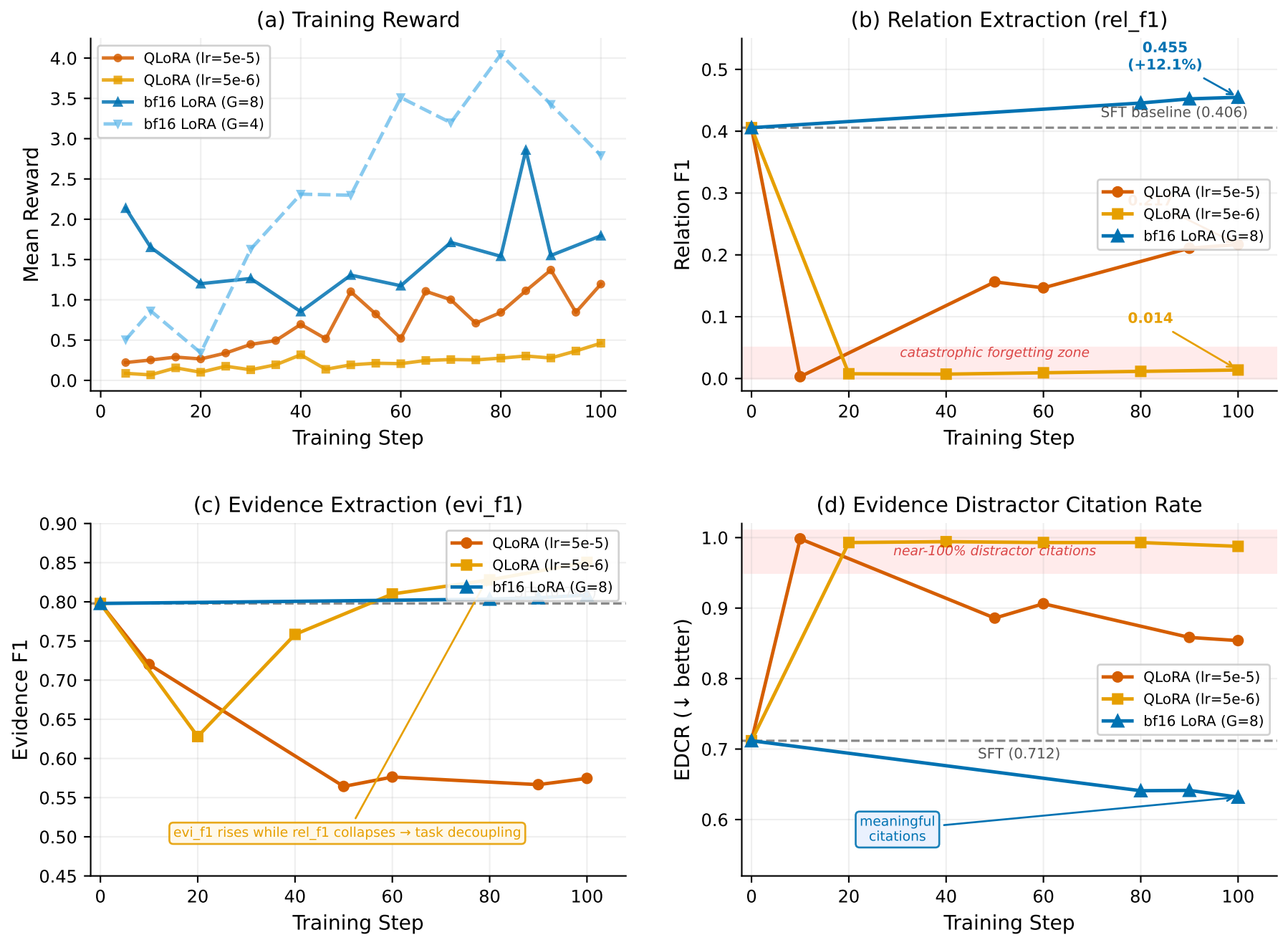
(a) Training Reward
Mean Reward
Training Step
QLoRA (lr=5e-5)
QLoRA (lr=5e-6)
bf16 LoRA (G=8)
bf16 LoRA (G=4)
(b) Relation Extraction (rel_f1)
Relation F1
Training Step
0.455
(+12.1%)
SFT baseline (0.406)
0.014
catastrophic forgetting zone
QLoRA (lr=5e-5)
QLoRA (lr=5e-6)
bf16 LoRA (G=8)
(c) Evidence Extraction (evi_f1)
Evidence F1
Training Step
evi_f1 rises while rel_f1 collapses → task decoupling
QLoRA (lr=5e-5)
QLoRA (lr=5e-6)
bf16 LoRA (G=8)
(d) Evidence Distractor Citation Rate
EDCR (↓ better)
Training Step
near-100% distractor citations
SFT (0.712)
meaningful
citations
QLoRA (lr=5e-5)
QLoRA (lr=5e-6)
bf16 LoRA (G=8)